

Azhagumurugan R

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Summary

Electronics and Communication Engineering undergraduate at Madras Institute of Technology, Anna University with strong foundations in C/C++, Embedded Systems, Digital Signal Processing, and Real-Time Systems. Experienced in FPGA-based system design, hardware-software integration, debugging, and system validation. Skilled in algorithm optimization, control systems fundamentals, and modular software development. Seeking a Software & Hardware Engineering role to contribute to industrial-grade embedded and product engineering systems.

Projects

RTL Design & Verification using Verilog

- Designed synthesizable RTL modules (ALU, FSM, counters) using Verilog/SystemVerilog with proper clocked logic implementation.
- Developed self-checking testbenches and performed functional and timing verification using ModelSim.

FPGA-Based Gun Shooting Detection & DOA System

- Developed FPGA-based real-time gunshot detection system using DSP techniques and multi-sensor data acquisition.
- Implemented TDOA-based Direction of Arrival (DOA) estimation with optimized latency for real-time performance.

Low-Cost CubeSat Attitude Determination and Control System (ADCS)

- Designed CubeSat ADCS using sensor fusion and PID-based feedback control for attitude stabilization.
- Simulated stabilization and disturbance rejection under dynamic operating conditions.

Graph-Based Financial Crime Detection Engine

- Built a modular graph-based fraud detection engine in Python,HTML Javascript,CSS,using DFS, BFS, and cycle detection algorithms.
- Optimized STL data structures for improved time complexity and memory efficiency.

Competitive Programming / C++ Problem Solving

- Solved 150+ Data Structures & Algorithms problems focusing on performance optimization and complexity analysis.
- Strong in graph algorithms, recursion, dynamic programming, and memory-efficient C++ implementation.

Skills

Programming Languages: C, C++, Python

Embedded & Hardware: Embedded Systems, FPGA Design, Verilog/SystemVerilog, Real-Time Systems, Hardware-Software Integration

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Algorithm Optimization, Debugging, SDLC

Core Engineering: Digital Electronics, Control Systems, Signal Processing, System Validation

Tools: Git, VS Code, ModelSim

Education

Madras Institute of Technology, Anna University

B.E. — Electronics & Communication Engineering (2024–2028 Expected)

Relevant Coursework: Digital Signal Processing, Control Systems, Embedded Systems, Data Structures & Algorithms

Achievements

- Finalist — Sprintathon'25 (National-Level Hackathon)
- Finalist — Pragyeen (National Institute of Technology) — National-Level Hackathon
- Finalist — RIFT — National-Level Hackathon
- 1st Prize — College-Level Technical Quiz Competition

Positions of Responsibility

- Treasurer — AUSEC (Anna University Entrepreneurship Club)
- Member — Raptors Drone Club